

# Bacteroidetes/Firmicutes ratio

Muluh

## 1 Bacteroidetes/Firmicutes ratio

### 1.1 Group-wise comparisons

- **Taxonomic level:** Phylum
- **Group variable:** diet
- **Multiple testing correction:** fdr
- **Statistical test** Wilcoxon

#### 1.1.1 Table of the top findings

#### 1.1.2 Across-group

##### 1.1.2.1 Before treatment

- Multiple testing correction: **fdr**
- Statistical test: **Wilcoxon**

Table 1: Table of Bacteroidetes/Firmicutes before Treatment

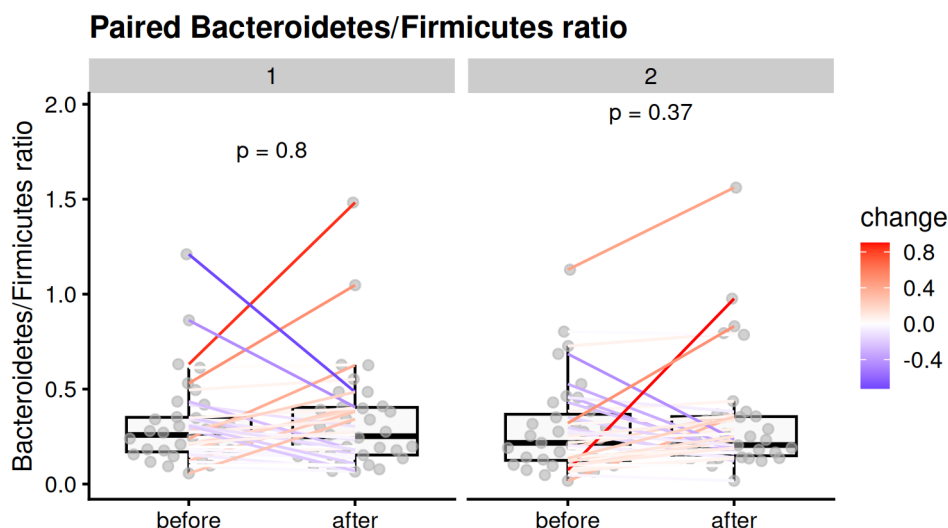
| group1 | group2 | mean1 | mean2 | log2FC | p.adj |
|--------|--------|-------|-------|--------|-------|
| 1      | 2      | 0.32  | 0.29  | -0.12  | 0.41  |

##### 1.1.2.2 After treatment

- Multiple testing correction: **fdr**
- Statistical test: **Wilcoxon**

Table 2: Table of Bacteroidetes/Firmicutes after Treatment

| group1 | group2 | mean1 | mean2 | log2FC | p.adj |
|--------|--------|-------|-------|--------|-------|
| 1      | 2      | 0.33  | 0.33  | -0.02  | 0.68  |



### 1.1.3 Within-group

#### 1.1.3.1 Paired treatment

- Multiple testing correction: **paired** -none
- Statistical test: **Wilcoxon**

Table 3: Table of Bacteroidetes/Firmicutes paired between timepoint (1 and 2)

| diet | mean1 | mean2 | log2FC | p.value |
|------|-------|-------|--------|---------|
| 1    | 0.32  | 0.33  | 0.28   | 0.80    |
| 2    | 0.29  | 0.33  | 1.07   | 0.37    |

Linear mixed model fit by REML. t-tests use Satterthwaite's method [ lmerModLmerTest]

Formula:  $y \sim \text{diet} * \text{time} + (1 | \text{id})$

Data: df\_no\_miss

REML criterion at convergence: 414

Scaled residuals:

|         |         |         |        |        |
|---------|---------|---------|--------|--------|
| Min     | 1Q      | Median  | 3Q     | Max    |
| -3.2394 | -0.4707 | -0.0571 | 0.5582 | 2.2699 |

Random effects:

| Groups   | Name        | Variance | Std.Dev. |
|----------|-------------|----------|----------|
| id       | (Intercept) | 0.5842   | 0.7643   |
| Residual |             | 0.6656   | 0.8159   |

Number of obs: 140, groups: id, 71

Fixed effects:

|             | Estimate | Std. Error | df        | t value | Pr(> t )     |
|-------------|----------|------------|-----------|---------|--------------|
| (Intercept) | -1.93048 | 0.33557    | 105.44036 | -5.753  | 8.67e-08 *** |
| diet2       | -0.50534 | 0.47123    | 105.43979 | -1.072  | 0.286        |
| time        | -0.02308 | 0.19712    | 68.20547  | -0.117  | 0.907        |
| diet2:time  | 0.22067  | 0.27679    | 68.19487  | 0.797   | 0.428        |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

```

Correlation of Fixed Effects:
              (Intr) diet2  time
diet2         -0.712
time          -0.875  0.623
diet2:time    0.623 -0.875 -0.712

```

## 2 Data Summary

```

class: TreeSummarizedExperiment
dim: 5 140
metadata(1): agglomerated_by_rank
assays(2): counts relabundance
rownames(5): Actinobacteria Bacteroidetes Firmicutes Other
              Proteobacteria
rowData names(10): kingdom phylum ... clade_name phylum_sub
colnames(140): AE1332 AK1371 ... TS2358 VK2336
colData names(16): sample id ... bact_firm_ratio bfratio
reducedDimNames(0):
mainExpName: NULL
altExpNames(0):
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL

```